Lab 3: Deformities in *C. elegans* intestine due to Infection

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Introduction

The aim of this experiment was to study the effect of bacterial infection on the morphology of *C. elegans* intestine. We used two different species of bacteria, *Escherichia coli* (strain OP50) and *Pseudomonas aeruginosa* (strain PA14). OP50 was the control, since it is the standard laboratory food for *C. elegans*. PA14 is known to be pathogenic to *C. elegans*. We infected L4 worms orally and observed the changes in their intestine after imaging with an epifluorescence microscope. We also performed a quantitative analysis of the intestinal morphology by calculating a deformation index for the intestine.

Materials and Methods

Materials

1. *C. elegans* master plate
2. 2 transfer plates with only agar
3. 3 *E. coli* plates (GFP tagged)
4. 3 *P. aeruginosa* plates (GFP tagged)
5. Parafilm
6. Two eyelash picks - made with eyelashes, toothpicks, 200 µL pipette tips, parafilm and fevicol
7. A dissecting microscope
8. 3% agar
9. Levamisole
10. Glass slides
11. Coverslips
12. An epifluorescence microscope
13. An incubator

Methods

Worm Picking

Using a master plate, we gently picked around 80 L4 worms using an eyelash pick and transferred them to a plain agar plate; 40 worms per plate. We left the worms on the plate for 20-30 minutes, so that the bacteria coating their body would be removed when they moved in the agar. The L4 worms were identifiable by a vulval opening. A dissection microscope was used to pick the worms.

Infection

We then transferred the worms to the experimental plates. For the control group, we transferred worms to three *E. coli* plates, 10 worms each. For the experimental group, we transferred worms to three *P. aeruginosa* plates, 10 worms each. We left the worms on the plates for 48 hours, at 25°C.

Imaging

The worm intestines had been tagged with mCherry, and both bacteria were tagged with GFP. On a glass slide, we dropped some 3% agar, to create a gel pad for the worms. The worms were anesthetized using levamisole, and a coverslip was placed on top. We imaged the worms using an epifluorescence microscope, and captured the images at a 40X magnification.

Results

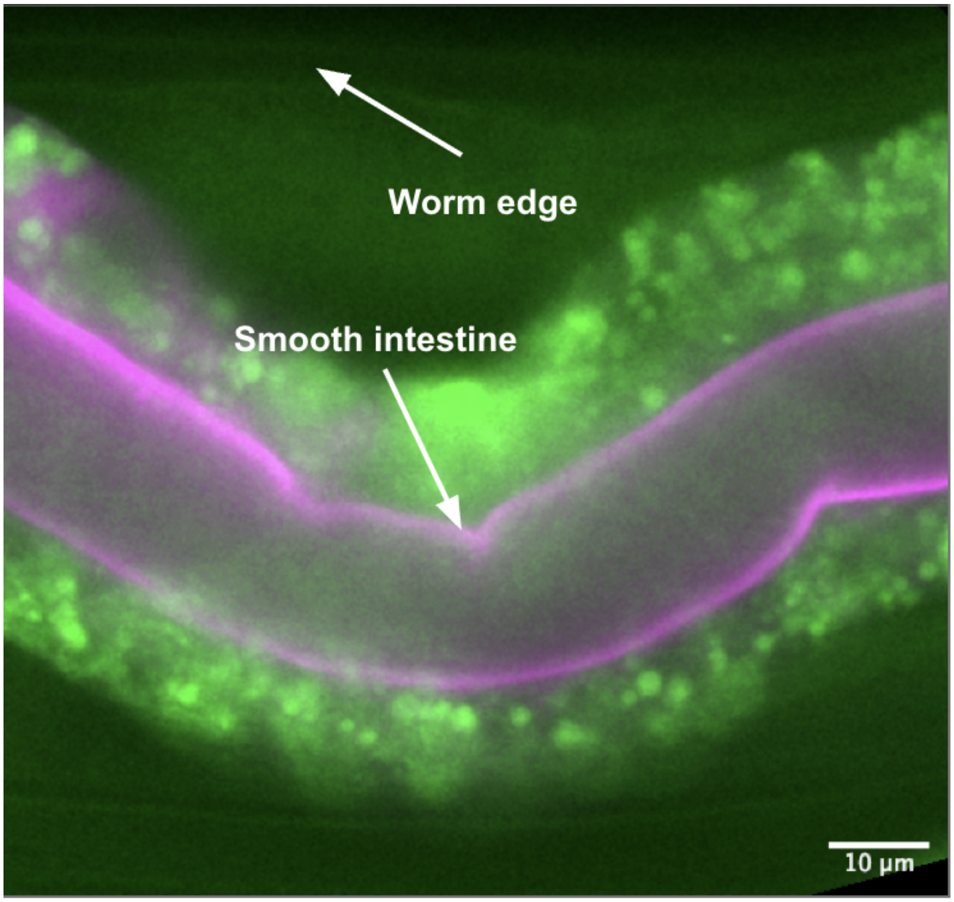

Figure 1: *E. coli* OP50 (control).

Figure 1 shows the section of a *C. elegans* intestine upon being fed *E. coli*, which is its standard food in the laboratory. The intestine of the worm is observed to be mostly smooth, without any kinks or inflammation.

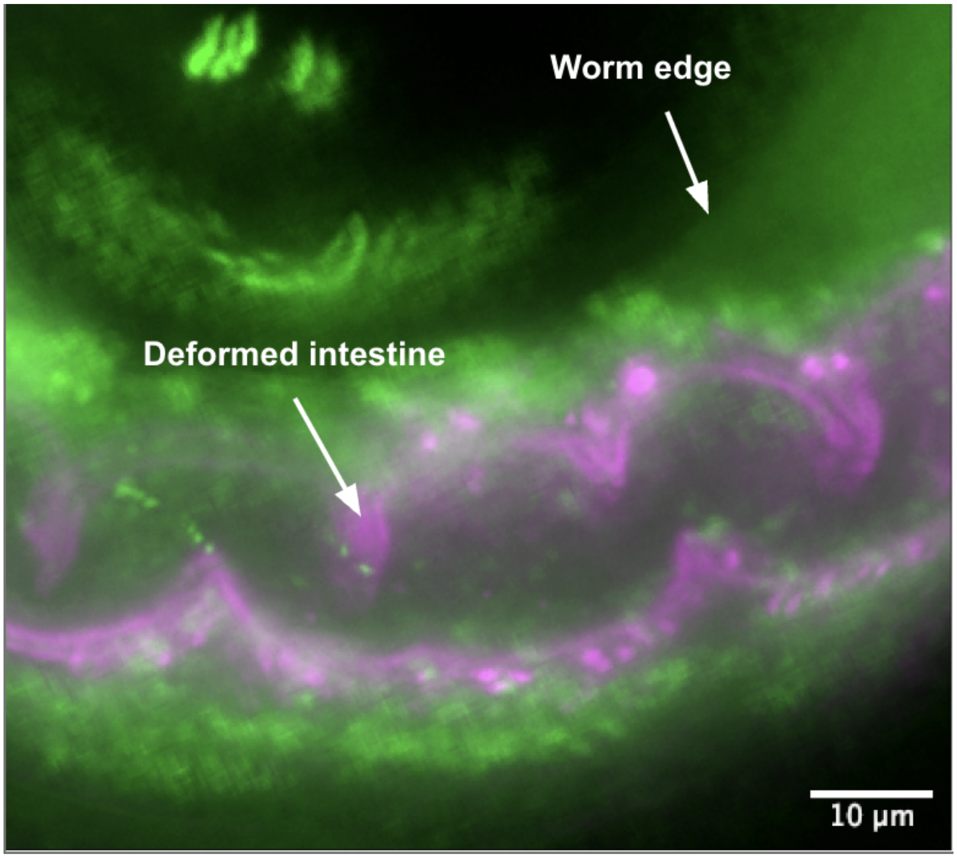

Figure 2: *P. aeruginosa* PA14 (infected).

Figure 2 shows the section of a *C. elegans* intestine upon being fed *P. aeruginosa*, which is a natural pathogen of the organism. The intestine appears to be more swollen, and the edges more deformed.

Deformation Index

To quantify the degree to which the worm had been affected, we defined a “deformation index”. This was calculated by taking the ratio of the length of the intestine to the length of the worm. In case of no or minimal deformation, the worm length should be approximately equal to the length of the intestine, giving a ratio of around 1. In case of high deformation, the length of the intestine will be much longer than the length of the worm, resulting in a ratio significantly greater than 1.

Analysis

The images were analyzed using Fiji. We used the segmented line tool to trace the length of the intestine and the worm, and then obtained the length in pixels using the measure command. Three sections of each worm had been imaged. The DI (deformation index) for each section was calculated, for the top and bottom respectively. This resulted in 6 DI values per worm. The average DI for each worm was then calculated, and is shown in the table below. Our sample size was 3 worms per bacterial exposure.

Table 1: Average Deformation Index per worm.

Bacteria	Average DI
OP50	1.0979
OP50	1.1424
OP50	1.1507
PA14	1.0865
PA14	1.5180
PA14	1.5740

Graph

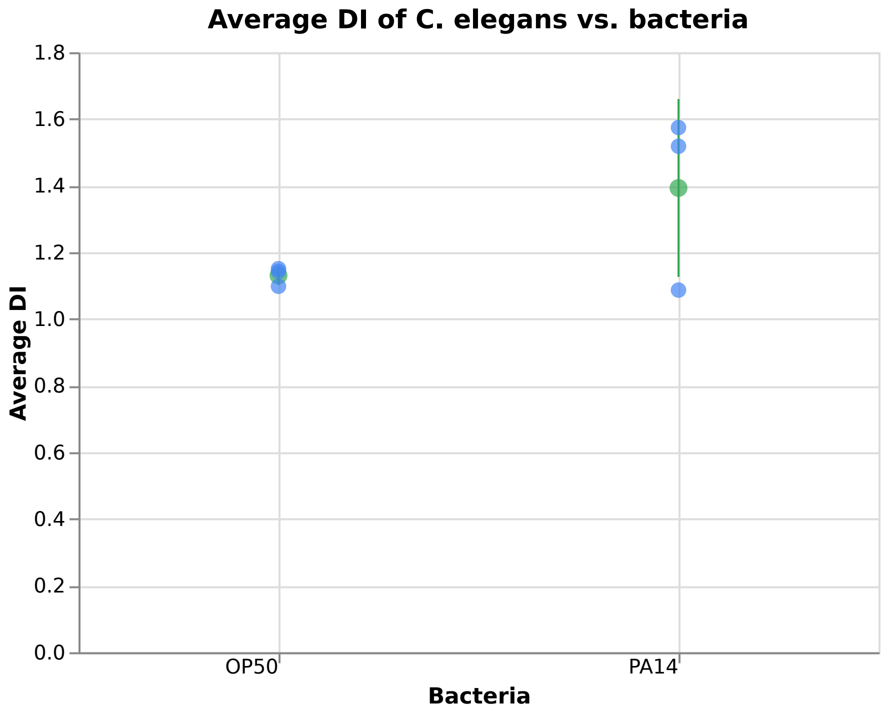

Figure 3: Average DI of *C. elegans* vs. bacteria type. Individual replicates shown in blue; green points and error bars represent the mean ± standard deviation.

Figure 3 shows the deformation indices of the worms. There are four values on the graph per infection: three of them denote the separate average DIs, and the green coordinate indicates the mean of the three average DIs per worm, with standard deviation error bars.

Statistical Significance

A one-tailed Welch's t-test was performed to determine whether there was a significant difference between the PA14 and OP50 infections. This test was chosen since the standard deviations of the two samples were unequal.

H_0: There is no significant difference between the DIs of the intestines of the worms upon being fed the different bacteria.

H_1: DI of PA14 worms > DI of OP50 worms

Significance Level: 0.05

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where $\bar{X}_1$ and $\bar{X}_2$ are the sample means, s_1 and s_2 are the sample standard deviations, and n_1 and n_2 are the sample sizes.

$$\therefore t = \frac{1.392838693 - 1.13033354}{\sqrt{\frac{0.2667850071^2}{3} + \frac{0.02840530039^2}{3}}}$$

$$t = 1.694685906 \approx 1.695$$

The degrees of freedom for Welch's t-test are calculated using the Welch-Satterthwaite equation:

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

$$\nu = \frac{\left(\frac{0.2667850071^2}{3} + \frac{0.02840530039^2}{3}\right)^2}{\frac{\left(\frac{0.2667850071^2}{3}\right)^2}{2} + \frac{\left(\frac{0.02840530039^2}{3}\right)^2}{2}}$$

$$\nu = 2.045 \approx 2$$

From the t-distribution table, the critical value for a one-tailed test with 2 degrees of freedom at a 0.05 significance level is 2.920.

i.e. $P(T > 2.920) = 0.05$

Hence, since $1.695 < 2.920, P(T > 1.695) > P(T > 2.920)$

So, $P(T > 1.695) > 0.05$

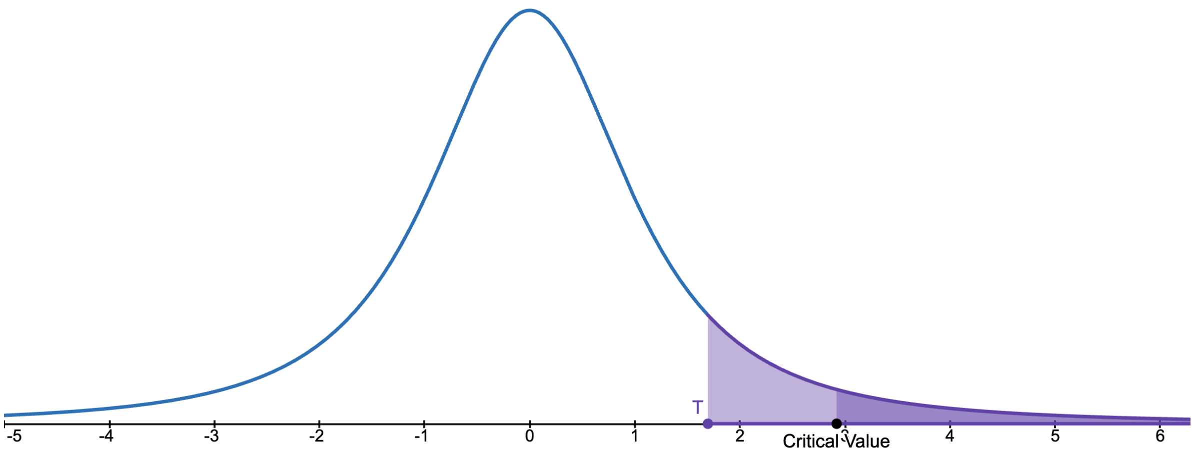

Figure 4: t-distribution with 2 degrees of freedom, plotted in Desmos.

So, we don't have enough evidence to reject the null hypothesis.

Discussion

Visibly, there does seem to be a difference in between the morphologies of the intestines of the two groups of worms. However, the statistical analysis does not support this observation. This could be due to a number of reasons.

Firstly, our sample size was too small, only 3 worms per group. So, it is highly likely that they do not represent the general population of worms under the two different infections. We can also see from the PA14 infection data, that there is one outlier, and the group's standard deviation is almost 10 times that of the OP50 group. This would have a huge effect in a small sample size.

We could have also picked incorrect worms. For example, if we had accidentally picked a mix of L3 and L4 worms, both would have reached the adult stage within 48 hours. Since it would be difficult to distinguish between the two adults, both would have been imaged, leading to inaccurate results.

Additional Notes

We were able to obtain only 2 OP50 fed worms and 1 PA14 fed worm from our batch. A lot of our worms died, and we had also accidentally picked some adults which reproduced, leading to a lot of L1s and L2s on the plates. Because of this, we had to use a few images from other batches.

In some of the images, we could see that the ingested bacteria was outside the intestine, for both OP50 and PA14. Upon further reading, this is possibly due to different reasons in the two groups. Intestinal barrier dysfunction is a common feature of aging across organisms. So it is likely that the OP50 worms with bacteria outside the intestine were simply old. In the PA14 worms, this was likely due to the infection itself, as PA14 is a pathogenic bacteria that can cause damage to the intestinal lining. (References linked below)

References

1. Cambridge International. *List of Formulae and Statistical Tables*. https://www.cambridgeinternational.org/Images/417318-list-of-formulae-and-statistical-tables.pdf
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3. Salazar, A.M., Aparicio, R., Clark, R.I., Rera, M. and Walker, D.W. (2023). Intestinal barrier dysfunction: an evolutionarily conserved hallmark of aging. *Disease Models & Mechanisms*, 16(4). https://doi.org/10.1242/dmm.049969
4. Irazoqui, J.E., Troemel, E.R., Feinbaum, R.L., Luhachack, L.G., Cezairliyan, B.O. and Ausubel, F.M. (2010). Distinct Pathogenesis and Host Responses during Infection of *C. elegans* by *P. aeruginosa* and *S. aureus*. *PLOS Pathogens*, 6(7), e1000982. https://doi.org/10.1371/journal.ppat.1000982